

Subham Ambastha

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Professional Summary

Computer Science undergraduate with project experience in software development, backend systems, machine learning, and full-stack application development. Proficient in Python and C++, with experience building scalable applications and solving real-world problems.

Education

Dayananda Sagar Academy of Technology & Management , Bengaluru Bachelor of Engineering in Computer Science	<i>Expected Jul 2027</i> CGPA: 8.98 / 10
Delhi Public School Bokaro , Jharkhand Higher Secondary Certificate (Science – PCM)	2022 89.33% (Top 10% of class)

Projects

Vaak – Full-Stack Language Assistant FastAPI MongoDB JWT REST APIs React - Developed a full-stack language assistant supporting multilingual translation, transliteration, dictionary lookup, and chat-based language assistance. - Designed and implemented secure authentication and user history management using JWT-based authorization and RESTful APIs. - Integrated external translation and dictionary APIs to provide real-time language detection, translations, word definitions, and example usage.
F1 Strategy Engineer – ML-Powered F1 Pit Strategy Simulator Python FastF1 XGBoost Streamlit Pandas - Developed an end-to-end Formula 1 strategy analysis platform using telemetry data from 33,569 racing laps across 23 Grand Prix events and 27 drivers. - Engineered an XGBoost-based tyre degradation and lap-time prediction model achieving an R^2 score of 0.973 on validation data. - Designed a custom undercut simulation engine that identifies crossover laps and evaluates pit-stop strategies through cumulative undercut advantage analysis.

Skills

Languages	C++, Python, JavaScript, SQL
Core CS	Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Computer Networks, Database Management Systems (DBMS), Operating Systems (OS)
Backend	FastAPI, REST APIs
Databases	MongoDB, MySQL
Machine Learning	NumPy, Pandas, XGBoost
Tools	Docker, Git

Certifications

- Microsoft – Fundamentals of Computer Networking • NPTEL – Foundations of Deep Learning • The Complete Git Guide
- Generative AI • JavaScript • HTML5 • CSS3